

Initiating Search

November 4, 2025, 4:01 PM

• Search:

CAS Lexicon Search:

Giese reaction - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (234)	 References	View Results
Exported: Retrieved Related Reaction Results (612)	 Reactions	View Results

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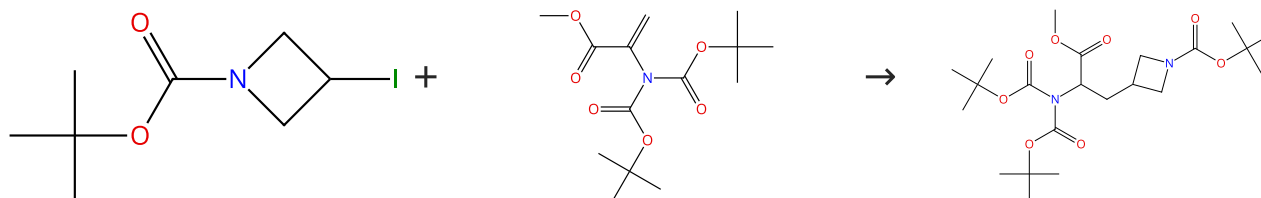
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Reactions (50)

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Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (93)

 Suppliers (48)

31-255-CAS-22353295

Steps: 1 Yield: 100%

Aminoalkyl radicals as halogen-atom transfer agents for activation of alkyl and aryl halides

1.1 Reagents: Triethylamine

Catalysts: 2,4,5,6-Tetra(9*H*-carbazol-9-yl)isophthalonitrile

Solvents: Acetonitrile, Water; 16 h, rt

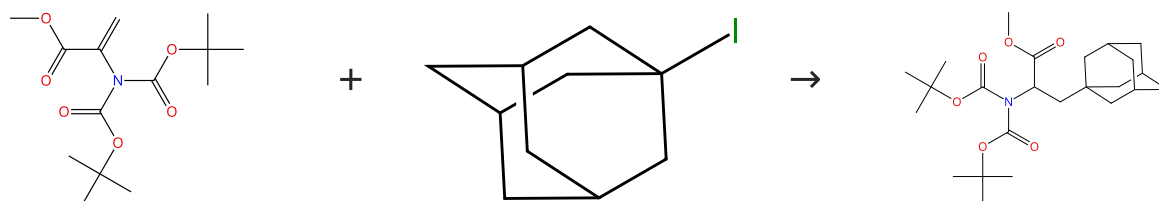
By: Constantin, Timothee; et al

Science (Washington, DC, United States) (2020), 367(6481), 1021-1026.

Experimental Protocols

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (48)

 Suppliers (56)

31-255-CAS-22353298

Steps: 1 Yield: 100%

Aminoalkyl radicals as halogen-atom transfer agents for activation of alkyl and aryl halides

1.1 Reagents: Triethylamine

Catalysts: 2,4,5,6-Tetra(9*H*-carbazol-9-yl)isophthalonitrile

Solvents: Acetonitrile, Water; 16 h, rt

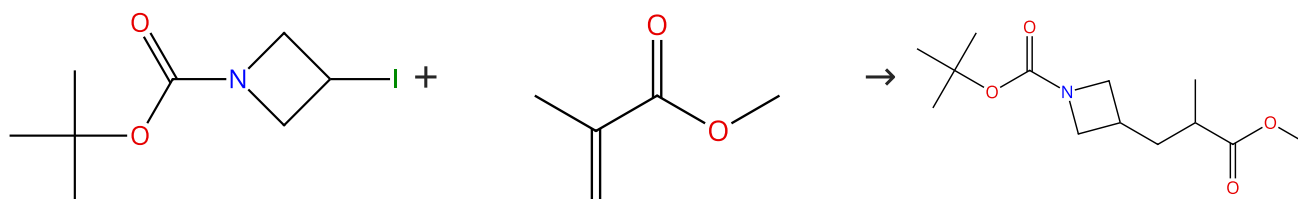
By: Constantin, Timothee; et al

Science (Washington, DC, United States) (2020), 367(6481), 1021-1026.

Experimental Protocols

Scheme 3 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (93)

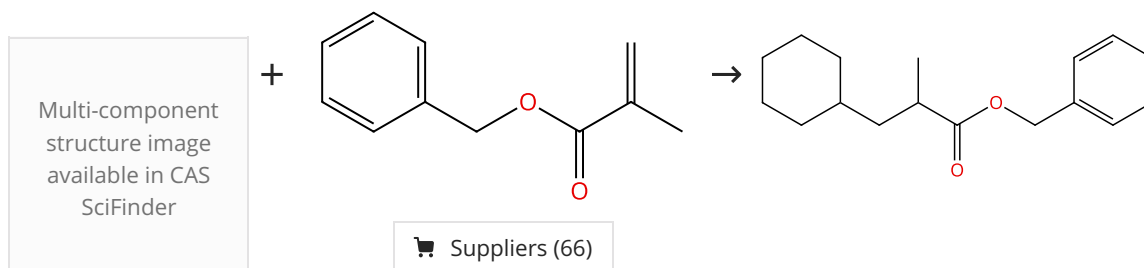
 Suppliers (78)

 Suppliers (3)

31-255-CAS-22353283	Steps: 1 Yield: 100%	Aminoalkyl radicals as halogen-atom transfer agents for activation of alkyl and aryl halides
1.1 Reagents: Triethylamine Catalysts: 2,4,5,6-Tetra(9 <i>H</i> -carbazol-9-yl)isophthalonitrile Solvents: Acetonitrile, Water; 16 h, rt		By: Constantin, Timothee; et al
Experimental Protocols		Science (Washington, DC, United States) (2020), 367(6481), 1021-1026.

Scheme 4 (1 Reaction)

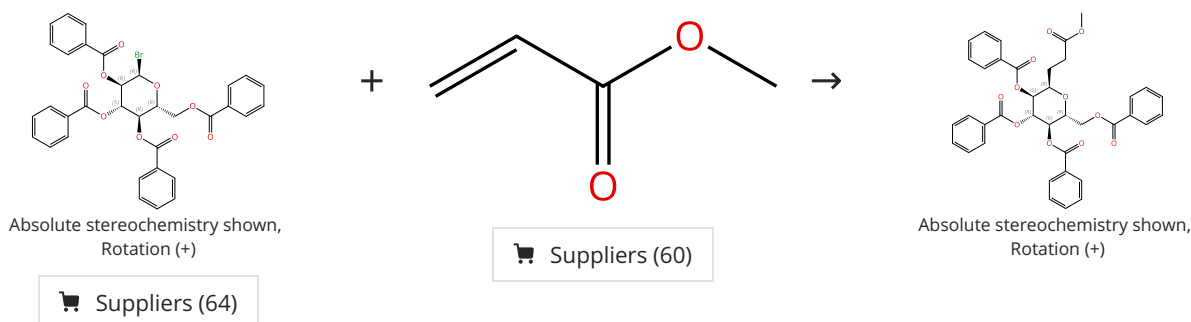
Steps: 1 Yield: 99%



31-255-CAS-23516535	Steps: 1 Yield: 99%	Photocatalytic Giese-Type Reaction with Alkylsilicates Bearing C,O-Bidentate Ligands
1.1 Reagents: Lithium hexafluorophosphate Catalysts: Acridinium, 3,6-bis(1,1-dimethylethyl)-10-phenyl-9-(2,4,6-trimethylphenyl)-, tetrafluoroborate(1-) (1:1) Solvents: Methanol, Acetone; 3 - 24 h, 23 °C		By: Morofuji, Tatsuya; et al
Experimental Protocols		Chemistry - A European Journal (2021), 27(22), 6713-6718.

Scheme 5 (1 Reaction)

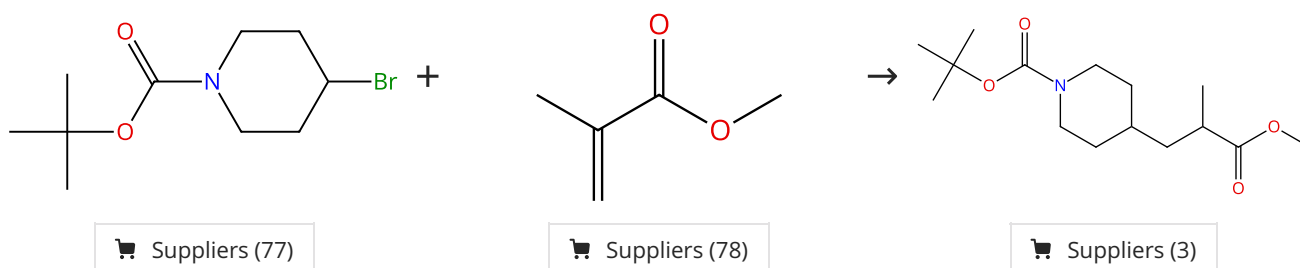
Steps: 1 Yield: 99%



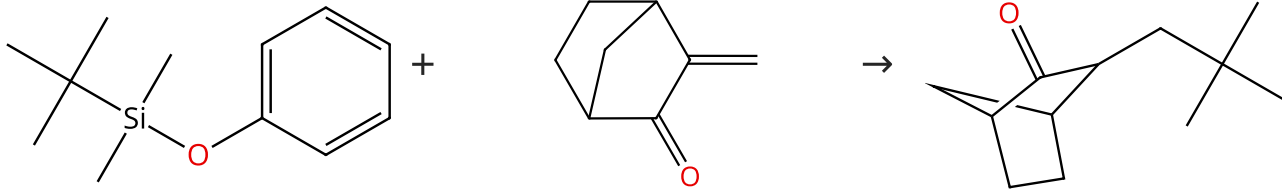
31-614-CAS-44059004	Steps: 1 Yield: 99%	Modulable photocatalyzed strategies for the synthesis of α-C-glycosyl alanine analogues via the Giese reaction with dehydroalanine derivatives
1.1 Reagents: Diisopropylethylamine Catalysts: Copper(1+), (2,9-dimethyl-4,7-diphenyl-1,10-phenanthroline-κ ^N ¹ ,κ ^N ¹⁰)[1,1'-(oxydi-2,1-phenylene)bis[1,1-diphenylphosphine-κ ^P]]-, (7-4)-, hexafluorophosphate(1-) (1:1) Solvents: Acetonitrile, Water; 2 h, rt		By: Poletti, Lorenzo; et al
Experimental Protocols		Organic Letters (2023), 25(26), 4862-4867.

Scheme 6 (1 Reaction)

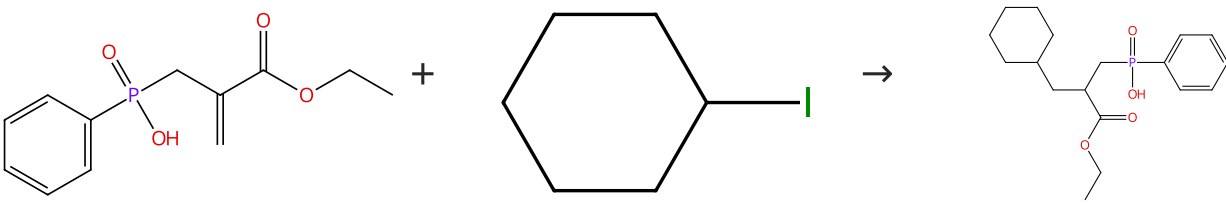
Steps: 1 Yield: 99%



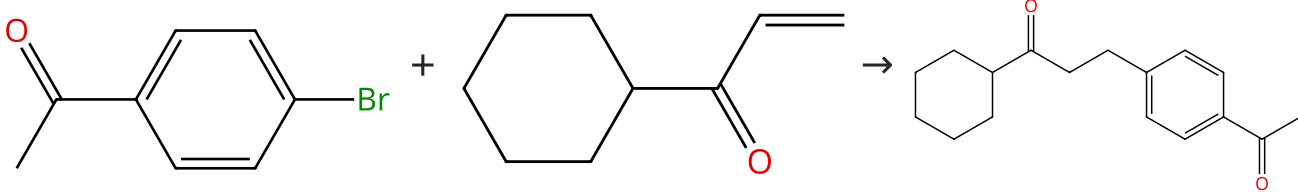
31-614-CAS-44055714 Steps: 1 Yield: 99% 1.1 Reagents: Cesium carbonate, Borate(1-), (carboxylato)(<i>N,N</i>-dimethylmethanamine)dihydro-, hydrogen, (7-4)- Catalysts: Benzonitrile, 2,3,4,5,6-penta-9<i>H</i>-carbazol-9-yl- Solvents: Dimethylformamide, Water; 12 h, rt Experimental Protocols	The Giese reaction of alkyl bromides using amine carboxy boranes By: Park, Changhee; et al Organic Chemistry Frontiers (2025), 12(6), 1874-1878.
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Scheme 7 (1 Reaction) Steps: 1 Yield: 98%   Suppliers (10)  Suppliers (51)  Suppliers (4)		
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31-614-CAS-42383421 Steps: 1 Yield: 98% 1.1 Catalysts: Biphenyl, Lithium perchlorate, Acridinium, 10-methyl-9-(2,4,6-trimethylphenyl)-, tetrafluoroborate(1-) (1:1) Solvents: Acetonitrile; 24 h, rt Experimental Protocols	Organophotocatalyzed C-Si Bond Fragmentation Using Silyl Ethers as Radical Precursors By: Luguera Ruiz, Adrian; et al Advanced Synthesis & Catalysis (2024), 366(19), 4132-4138.
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Scheme 8 (1 Reaction) Steps: 1 Yield: 98%   Suppliers (72)		
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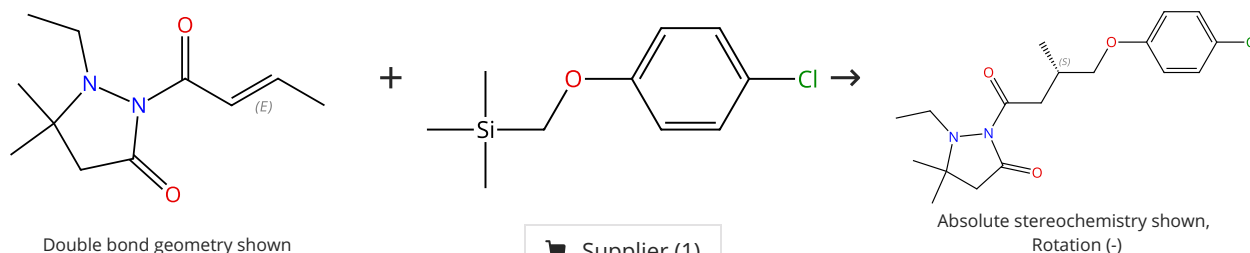
31-255-CAS-20187150 Steps: 1 Yield: 98% 1.1 Reagents: Tributylstannane Catalysts: Triethylborane Solvents: Dichloromethane, Hexane; 5 min, 0 °C; 30 min, 0 °C; 30 min, 0 °C 1.2 Reagents: Hydrochloric acid Solvents: Water; 10 min, 0 °C 1.3 Reagents: Hydrochloric acid Solvents: Ethyl acetate, Water; 0 °C	Late-stage diversification of phosphinic dehydroalanine pseudopeptides based on a Giese-type radical C-alkylation strategy By: Voreakos, Kostas; et al Organic Letters (2019), 21(12), 4397-4401.
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Scheme 9 (1 Reaction) Steps: 1 Yield: 97%   Suppliers (84)  Suppliers (23)		
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31-614-CAS-36776330 Steps: 1 Yield: 97% 1.1 Reagents: Hantzsch ester Catalysts: Dibromo[1,2-di(methoxy-κO)ethane]nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine, Iridium(1+), [4,4'-bis(1,1-dimethylethyl)-2,2'-bipyridine-$\kappa N^1, \kappa N^1$]bis[2-(2-pyridinyl-κM)phenyl-κC]-, (OC-6-33)-, hexafluorophosphate(1-) (1:1) Solvents: Acetonitrile; 15 min, 20 - 30 °C 1.2 24 h, 20 - 30 °C Experimental Protocols	Photoexcited Ni^{II}-Aryl Complex-Mediated Giese Reaction of Aryl Bromides By: He, Xian-Chen; et al Organic Letters (2023), 25(22), 4056-4060.
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Scheme 10 (1 Reaction)

Steps: 1 Yield: 96%

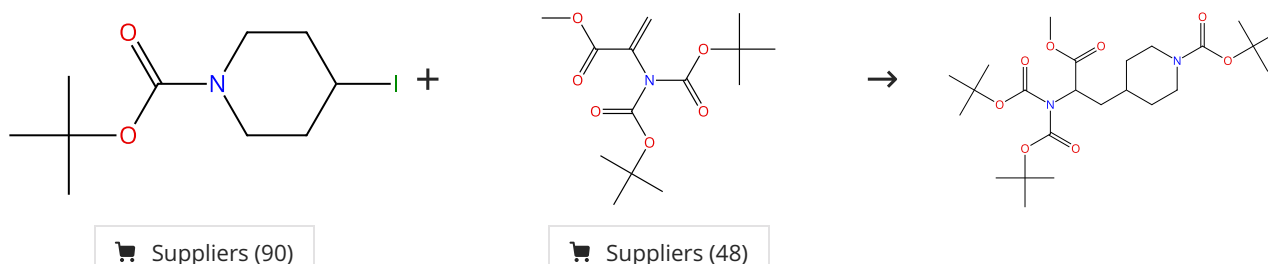


Suppliers (5)

31-614-CAS-27641927 Steps: 1 Yield: 96% 1.1 Catalysts: Tetrabutylammonium chloride, 2,6-Bis[(4S)-4,5-dihydro-4-[(1S)-1-methylpropyl]-2-oxazolyl]pyridine, Scandium triflate, Iridium(1+), [4,4'-bis(1,1-dimethylethyl)-2,2'-bipyridine-$\kappa N^1, \kappa N^1$]bis[3,5-difluoro-2-[5-(trifluoromethyl)-2-pyridinyl-κM]phenyl-κC]-, (OC-6-33)-, hexafluorophosphate(1-) (1:1) Solvents: Acetonitrile; rt → -25 °C; 24 h, -25 °C Experimental Protocols	Enantioselective Synthesis of γ-Oxycarbonyl Motifs by Conjugate Addition of Photogenerated α-Alkoxy Radicals By: Dong, Xiao; et al Organic Letters (2021), 23(15), 5703-5708.
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Scheme 11 (1 Reaction)

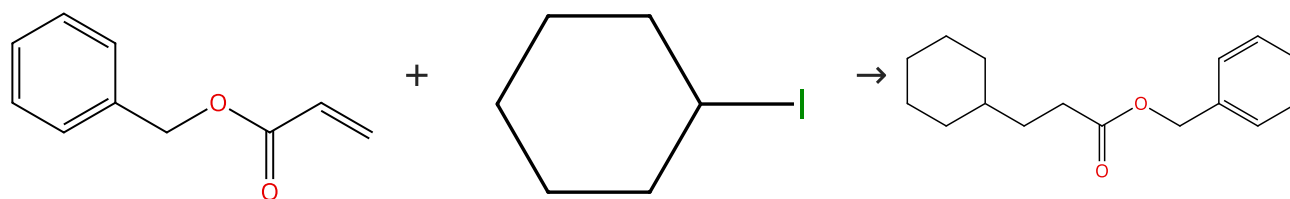
Steps: 1 Yield: 96%



31-255-CAS-22353297 Steps: 1 Yield: 96% 1.1 Reagents: Triethylamine Catalysts: 2,4,5,6-Tetra(9H-carbazol-9-yl)isophthalonitrile Solvents: Acetonitrile, Water; 16 h, rt Experimental Protocols	Aminoalkyl radicals as halogen-atom transfer agents for activation of alkyl and aryl halides By: Constantin, Timothee; et al Science (Washington, DC, United States) (2020), 367(6481), 1021-1026.
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Scheme 12 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (73)

Suppliers (72)

Suppliers (2)

31-255-CAS-20578138

Steps: 1 Yield: 96%

Visible-light-initiated manganese-catalyzed Giese addition of unactivated alkyl iodides to electron-poor olefins

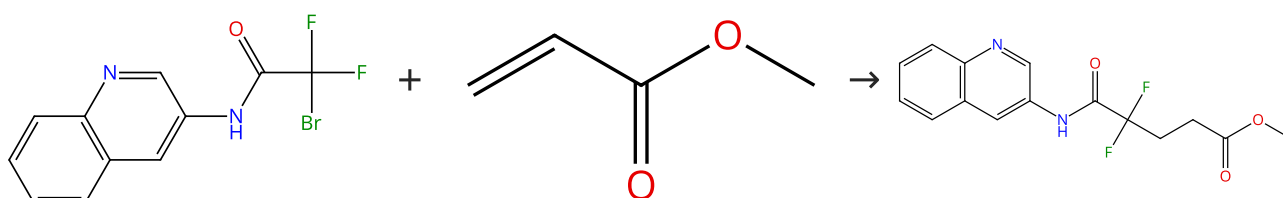
By: Dong, Jianyang; et al

Chemical Communications (Cambridge, United Kingdom) (2019), 55(78), 11707-11710.

1.1 **Reagents:** Hantzsch ester
Catalysts: Dimanganese decacarbonyl
Solvents: Dimethyl sulfoxide; 24 h, rt

Scheme 13 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (60)

31-614-CAS-41533731

Steps: 1 Yield: 95%

Metal-Free Visible-Light-Induced Hydrodifluoroalkylation of Electron-Deficient Alkenes Using Hantzsch Ester

By: Wang, Juan; et al

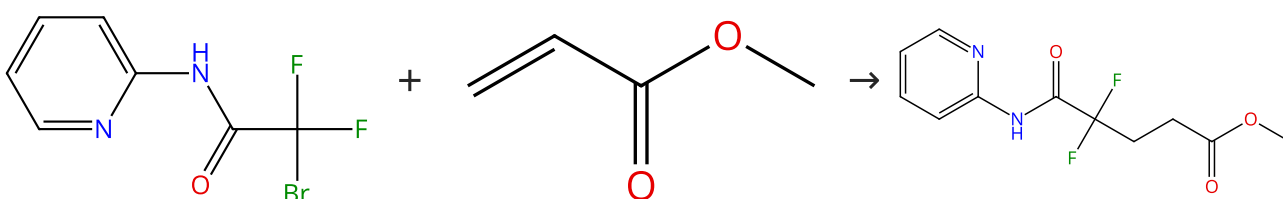
European Journal of Organic Chemistry (2024), 27(38), e202400591.

1.1 **Reagents:** Triethylamine, Hantzsch ester
Solvents: 1,4-Dioxane; 12 h, rt

Experimental Protocols

Scheme 14 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (3)

Suppliers (60)

31-614-CAS-41533739

Steps: 1 Yield: 95%

Metal-Free Visible-Light-Induced Hydrodifluoroalkylation of Electron-Deficient Alkenes Using Hantzsch Ester

By: Wang, Juan; et al

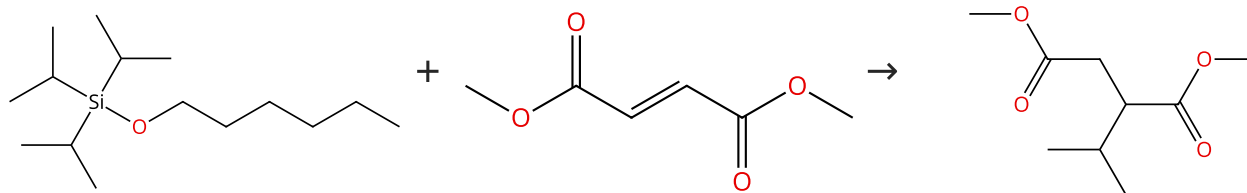
European Journal of Organic Chemistry (2024), 27(38), e202400591.

1.1 **Reagents:** Triethylamine, Hantzsch ester
Solvents: 1,4-Dioxane; 12 h, rt

Experimental Protocols

Scheme 15 (1 Reaction)

Steps: 1 Yield: 95%



Supplier (1)

Suppliers (23)

Suppliers (3)

31-614-CAS-42383427

Steps: 1 Yield: 95%

Organophotocatalyzed C-Si Bond Fragmentation Using Silyl Ethers as Radical Precursors

By: Luguera Ruiz, Adrian; et al

Advanced Synthesis & Catalysis (2024), 366(19), 4132-4138.

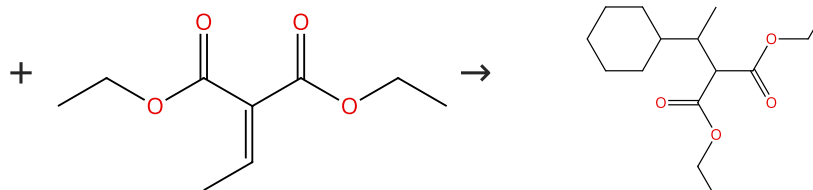
1.1 **Catalysts:** Biphenyl, Lithium perchlorate, Acridinium, 10-methyl-9-(2,4,6-trimethylphenyl)-, tetrafluoroborate(1-) (1:1)
Solvents: Acetonitrile; 24 h, rt

Experimental Protocols

Scheme 16 (1 Reaction)

Steps: 1 Yield: 95%

Multi-component
structure image
available in CAS
SciFinder



Suppliers (66)

Supplier (1)

31-255-CAS-23514628

Steps: 1 Yield: 95%

Photocatalytic Giese-Type Reaction with Alkylsilicates Bearing C,O-Bidentate Ligands

By: Morofuji, Tatsuya; et al

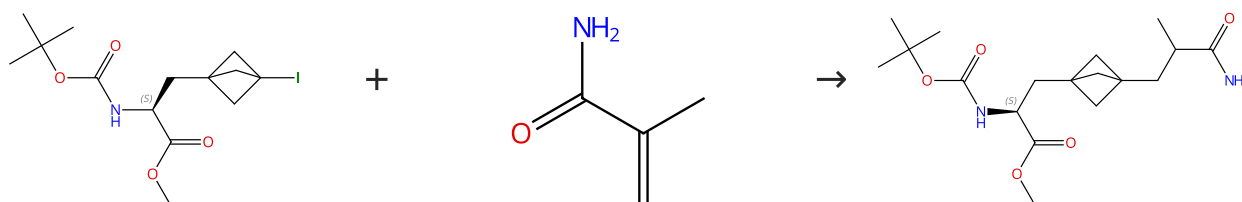
Chemistry - A European Journal (2021), 27(22), 6713-6718.

1.1 **Catalysts:** Acridinium, 3,6-bis(1,1-dimethylethyl)-10-phenyl-9-(2,4,6-trimethylphenyl)-, tetrafluoroborate(1-) (1:1)
Solvents: Methanol, Acetone; 3 - 24 h, 23 °C

Experimental Protocols

Scheme 17 (1 Reaction)

Steps: 1 Yield: 95%



Absolute stereochemistry shown

DNA conjugated

Absolute stereochemistry shown

DNA conjugated

Suppliers (4)

31-614-CAS-33433004

Steps: 1 Yield: 95%

On-DNA Hydroalkylation to Introduce Diverse Bicyclo[1.1.1]pentanes and Abundant Alkyls via Halogen Atom Transfer

By: Yen-Pon, Expedite; et al

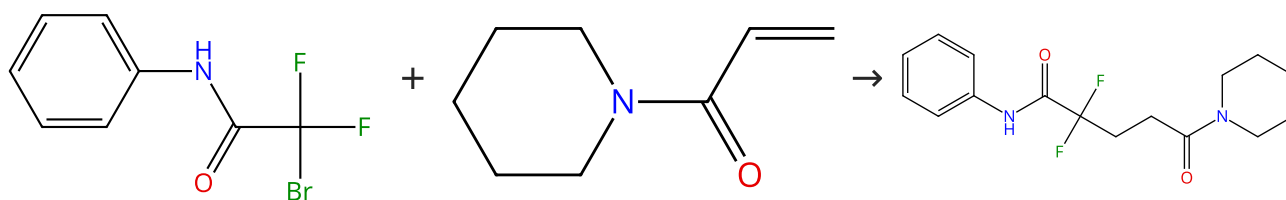
Journal of the American Chemical Society (2022), 144(27), 12184-12191.

1.1 **Reagents:** Sodium carbonate, 1,1,1,3,3,3-Hexamethyl-2-(trimethylsilyl)-2-trisilanol
Catalysts: 2,4,5,6-Tetra(9H-carbazol-9-yl)isophthalonitrile
Solvents: Dimethyl sulfoxide, Water; 5 min, rt

Experimental Protocols

Scheme 18 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (26)

Suppliers (44)

31-614-CAS-41533742

Steps: 1 Yield: 95%

Metal-Free Visible-Light-Induced Hydrodifluoroalkylation of Electron-Deficient Alkenes Using Hantzsch Ester

By: Wang, Juan; et al

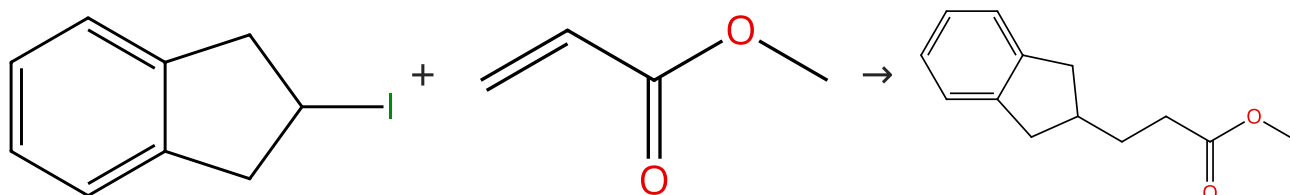
European Journal of Organic Chemistry (2024), 27(38), e202400591.

 1.1 Reagents: Triethylamine, Hantzsch ester
 Solvents: 1,4-Dioxane; 12 h, rt

Experimental Protocols

Scheme 19 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (43)

Suppliers (60)

Suppliers (4)

31-255-CAS-20578162

Steps: 1 Yield: 95%

Visible-light-initiated manganese-catalyzed Giese addition of unactivated alkyl iodides to electron-poor olefins

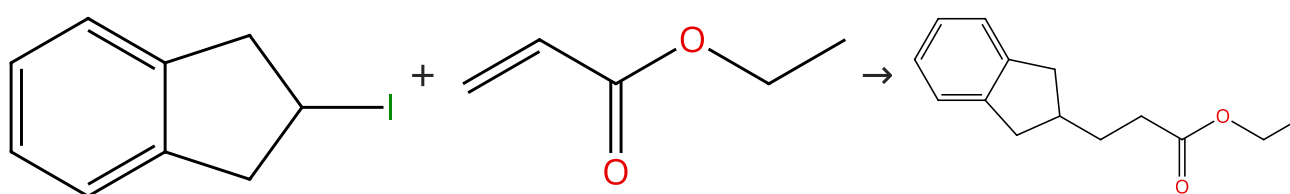
By: Dong, Jianyang; et al

Chemical Communications (Cambridge, United Kingdom) (2019), 55(78), 11707-11710.

 1.1 Reagents: Hantzsch ester
 Catalysts: Dimanganese decacarbonyl
 Solvents: Dimethyl sulfoxide; 24 h, rt

Scheme 20 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (43)

Suppliers (76)

Suppliers (5)

31-255-CAS-20578163

Steps: 1 Yield: 95%

Visible-light-initiated manganese-catalyzed Giese addition of unactivated alkyl iodides to electron-poor olefins

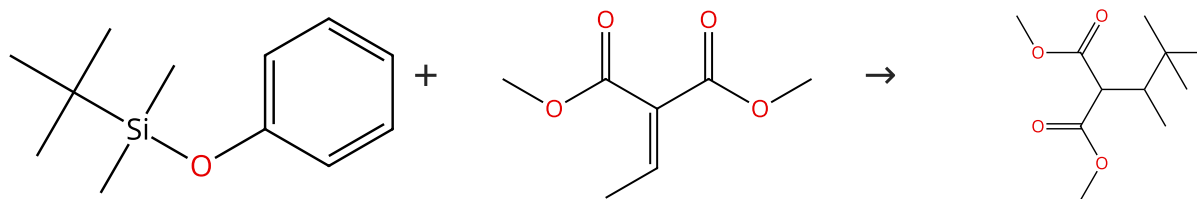
By: Dong, Jianyang; et al

Chemical Communications (Cambridge, United Kingdom) (2019), 55(78), 11707-11710.

 1.1 Reagents: Hantzsch ester
 Catalysts: Dimanganese decacarbonyl
 Solvents: Dimethyl sulfoxide; 24 h, rt

Scheme 21 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (10)

Suppliers (49)

31-614-CAS-42383404

Steps: 1 Yield: 95%

Organophotocatalyzed C-Si Bond Fragmentation Using Silyl Ethers as Radical Precursors

By: Luguera Ruiz, Adrian; et al

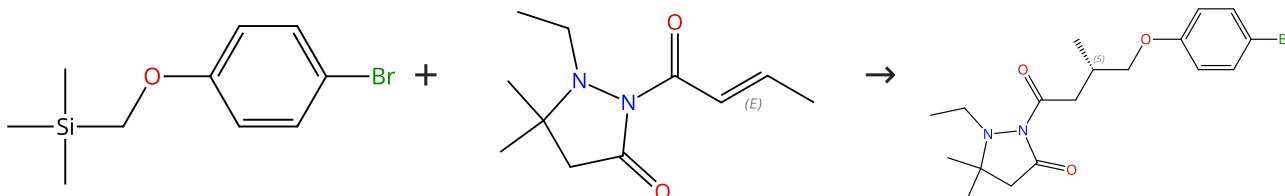
Advanced Synthesis & Catalysis (2024), 366(19), 4132-4138.

1.1 **Catalysts:** Biphenyl, Lithium perchlorate, Acridinium, 10-methyl-9-(2,4,6-trimethylphenyl)-, tetrafluoroborate(1-) (1:1)
Solvents: Acetonitrile; 24 h, rt

Experimental Protocols

Scheme 22 (1 Reaction)

Steps: 1 Yield: 94%



Double bond geometry shown

Absolute stereochemistry shown,
Rotation (-)

Suppliers (5)

31-614-CAS-27641934

Steps: 1 Yield: 94%

Enantioselective Synthesis of γ -Oxycarbonyl Motifs by Conjugate Addition of Photogenerated α -Alkoxy Radicals

By: Dong, Xiao; et al

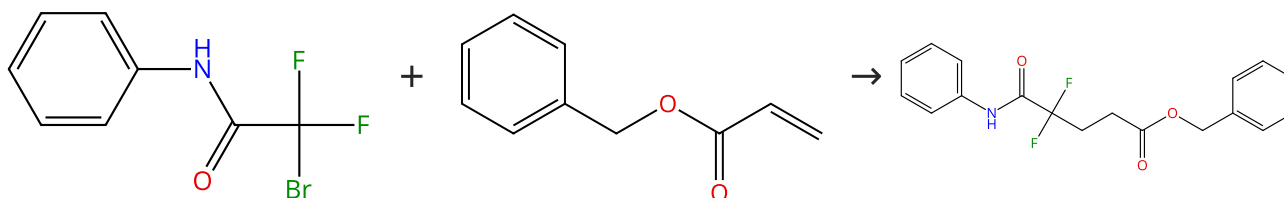
Organic Letters (2021), 23(15), 5703-5708.

1.1 **Catalysts:** Tetrabutylammonium chloride, 2,6-Bis[(4*S*)-1-methylpropyl]-2-oxazoly]pyridine, Scandium triflate, Iridium(1+), [4,4'-bis(1,1-dimethylethyl)-2,2'-bipyridine- κ^N , κ^N]bis[3,5-difluoro-2-[5-(trifluoromethyl)-2-pyridinyl- κ^N]phenyl- κ^C]-, (OC-6-33)-, hexafluorophosphate(1-) (1:1)
Solvents: Acetonitrile; rt $\rightarrow$ -25 °C; 24 h, -25 °C

Experimental Protocols

Scheme 23 (1 Reaction)

Steps: 1 Yield: 94%



Suppliers (26)

Suppliers (73)

31-614-CAS-41533733

Steps: 1 Yield: 94%

Metal-Free Visible-Light-Induced Hydrodifluoroalkylation of Electron-Deficient Alkenes Using Hantzsch Ester

By: Wang, Juan; et al

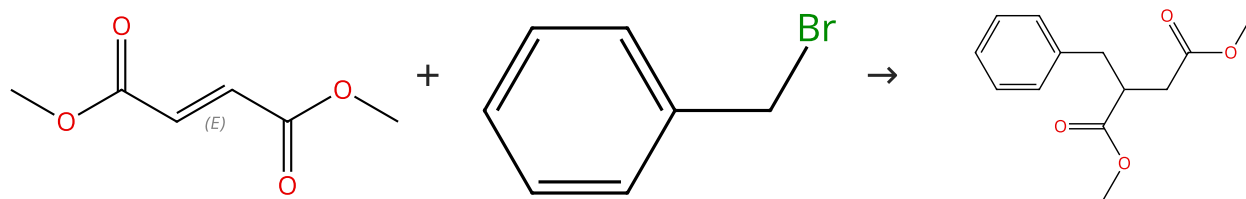
European Journal of Organic Chemistry (2024), 27(38), e202400591.

1.1 **Reagents:** Triethylamine, Hantzsch ester
Solvents: 1,4-Dioxane; 12 h, rt

Experimental Protocols

Scheme 24 (1 Reaction)

Steps: 1 Yield: 94%



Double bond geometry shown

Suppliers (111)

Suppliers (67)

Suppliers (9)

31-255-CAS-20099074

Steps: 1 Yield: 94%

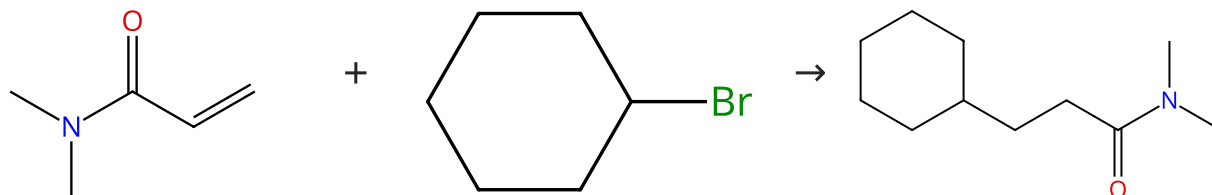
Photochemical generation of radicals from alkyl electrophiles using a nucleophilic organic catalyst

1.1 **Reagents:** γ -Terpinene, 2,6-Lutidine
Catalysts: 1*H*-Indole-1-carbodithioic acid, 5-bromo-, potassium salt (1:1)
Solvents: Acetonitrile; 24 h, 60 °C

By: Schweitzer-Chaput, Bertrand; et al
 Nature Chemistry (2019), 11(2), 129-135.

Scheme 25 (1 Reaction)

Steps: 1 Yield: 93%



Suppliers (68)

Suppliers (74)

Suppliers (8)

31-614-CAS-44585154

Steps: 1 Yield: 93%

MeOH-Triggered Halogen-Atom Transfer of Unactivated Alkyl Bromides Enabling the Photoredox Giese Addition

1.1 **Reagents:** Methanol, Cesium carbonate
Catalysts: Dibromo[1,2-di(methoxy- κ O)ethane]nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine, Iridium(1+), [4,4'-bis(1,1-dimethylethyl)-2,2'-bipyridine- κ N¹, κ N^{1'}]-bis[3,5-difluoro-2-[5-(trifluoromethyl)-2-pyridinyl- κ M]phenyl- κ C]-, (OC-6-33)-, hexafluorophosphate(1-) (1:1); 15 min, rt

By: He, Xian-Chen; et al

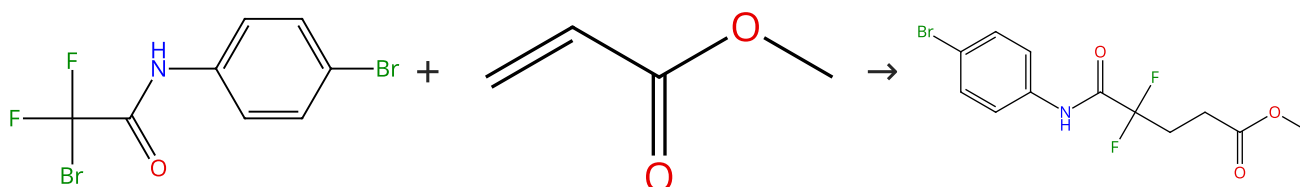
Organic Letters (2025), 27(12), 3089-3094.

1.2 24 h, rt

Experimental Protocols

Scheme 26 (1 Reaction)

Steps: 1 Yield: 93%



Suppliers (25)

Suppliers (60)

31-614-CAS-41533734

Steps: 1 Yield: 93%

Metal-Free Visible-Light-Induced Hydrodifluoroalkylation of Electron-Deficient Alkenes Using Hantzsch Ester

1.1 **Reagents:** Triethylamine, Hantzsch ester
Solvents: 1,4-Dioxane; 12 h, rt

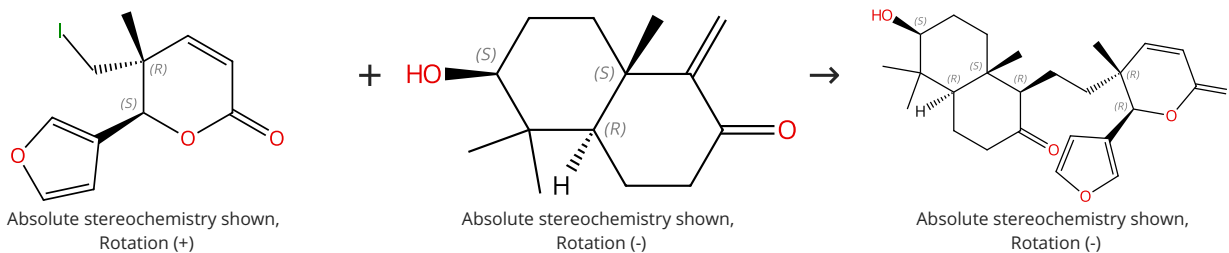
By: Wang, Juan; et al

European Journal of Organic Chemistry (2024), 27(38), e202400591.

Experimental Protocols

Scheme 27 (1 Reaction)

Steps: 1 Yield: 93%



31-614-CAS-35663448

Steps: 1 Yield: 93%

Concise Chemoenzymatic Synthesis of Gedunin

1.1 Reagents: Pyridine, Zinc, Cuprous iodide
Solvents: Water; rt; 4 h, 45 °C

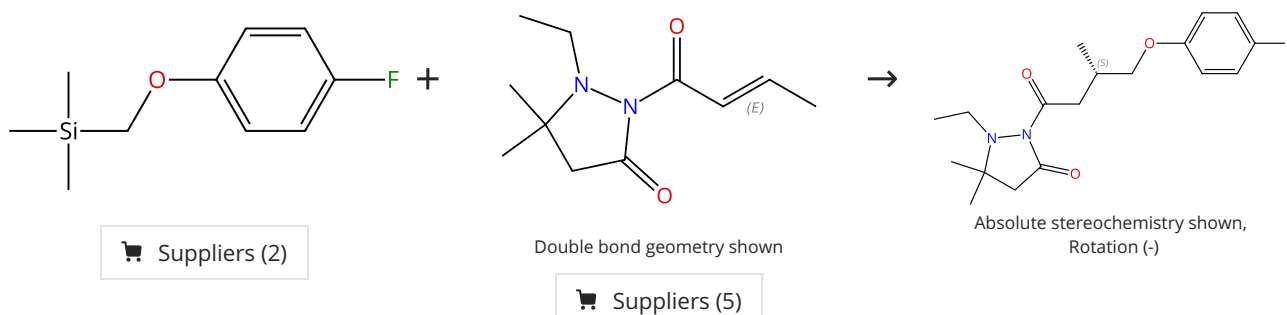
By: Li, Jian; et al

Journal of the American Chemical Society (2022), 144(42), 19238-19242.

Experimental Protocols

Scheme 28 (1 Reaction)

Steps: 1 Yield: 93%



31-614-CAS-27641923

Steps: 1 Yield: 93%

Enantioselective Synthesis of γ -Oxycarbonyl Motifs by Conjugate Addition of Photogenerated α -Alkoxy Radicals

1.1 Catalysts: Tetrabutylammonium chloride, 2,6-Bis[(4*S*)-4,5-dihydro-4-[(1*S*)-1-methylpropyl]-2-oxazolyl]pyridine, Scandium triflate, Iridium(1+), [4,4'-bis(1,1-dimethylethyl)-2,2'-bipyridine- $\kappa N^1, \kappa N^1$]bis[3,5-difluoro-2-[5-(trifluoromethyl)-2-pyridinyl- κN]phenyl- κC]-, (OC-6-33)-, hexafluorophosphate(1-)(1:1)

By: Dong, Xiao; et al

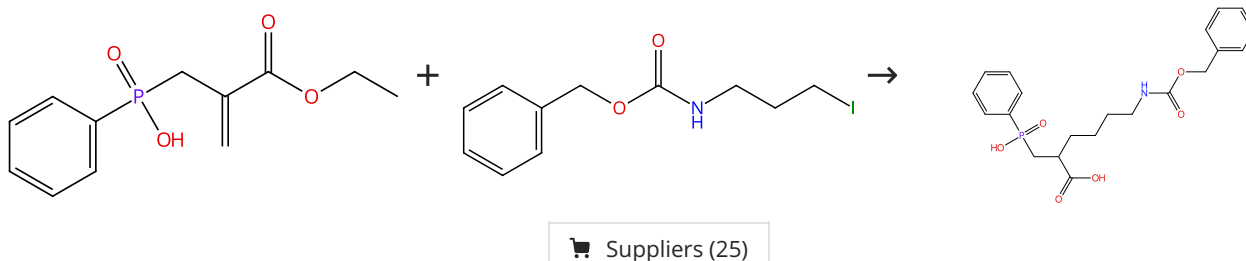
Organic Letters (2021), 23(15), 5703-5708.

Solvents: Acetonitrile; rt $\rightarrow$ -25 °C; 24 h, -25 °C

Experimental Protocols

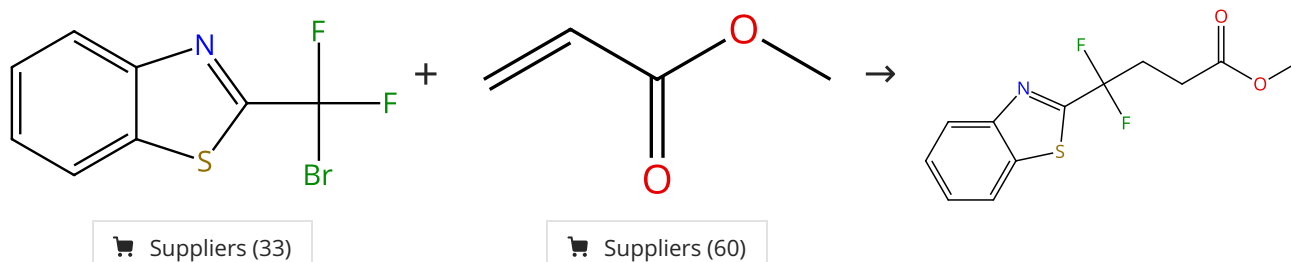
Scheme 29 (1 Reaction)

Steps: 1 Yield: 92%



31-255-CAS-20187114	Steps: 1 Yield: 92%	Late-stage diversification of phosphinic dehydroalanine pseudopeptides based on a Giese-type radical C-alkylation strategy
1.1 Reagents: Tris(trimethylsilyl)silane Catalysts: Azobisisobutyronitrile Solvents: Toluene; 1 h, 80 °C; 80 °C → rt		By: Voreakos, Kostas; et al
1.2 Reagents: Hydrochloric acid Solvents: Ethyl acetate, Water; rt		Organic Letters (2019), 21(12), 4397-4401.
1.3 -		
1.4 Reagents: Sodium hydroxide Solvents: Ethanol, Water; 24 h, rt		
1.5 Reagents: Hydrochloric acid Solvents: Water; pH 1, rt		

Scheme 30 (1 Reaction)

Steps: **1** Yield: **92%**

31-614-CAS-41533760	Steps: 1 Yield: 92%	Metal-Free Visible-Light-Induced Hydrodifluoroalkylation of Electron-Deficient Alkenes Using Hantzsch Ester By: Wang, Juan; et al European Journal of Organic Chemistry (2024), 27(38), e202400591.
1.1 Reagents: Triethylamine, Hantzsch ester Solvents: 1,4-Dioxane; 12 h, rt		
Experimental Protocols		

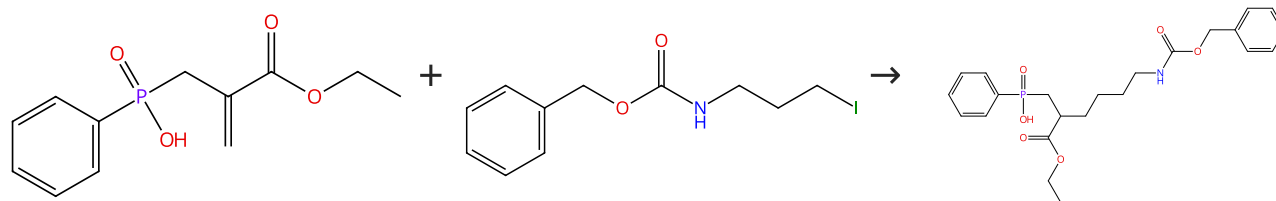
Scheme 31 (1 Reaction)

Steps: **1** Yield: **92%**

31-255-CAS-20187140	Late-stage diversification of phosphinic dehydroalanine pseudopeptides based on a Giese-type radical C-alkylation strategy
Steps: 1 Yield: 92%	
1.1 Reagents: Sodium cyanoborohydride Catalysts: Azobisisobutyronitrile Solvents: Methanol; 0 °C; 1 h, 80 °C; 80 °C → rt	By: Voreakos, Kostas; et al
1.2 Reagents: Hydrochloric acid Solvents: Ethyl acetate, Water; rt	Organic Letters (2019), 21(12), 4397-4401.

Scheme 32 (1 Reaction)

Steps: 1 Yield: 92%



Suppliers (25)

31-255-CAS-20187154

Steps: 1 Yield: 92%

Late-stage diversification of phosphinic dehydroalanine pseudopeptides based on a Giese-type radical C-alkylation strategy

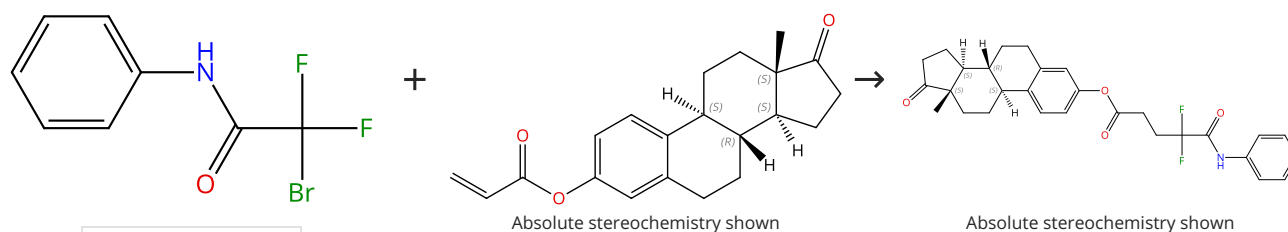
By: Voreakos, Kostas; et al

Organic Letters (2019), 21(12), 4397-4401.

- 1.1 **Reagents:** Tris(trimethylsilyl)silane
Catalysts: Azobisisobutyronitrile
Solvents: Toluene; 0 °C; 1 h, 80 °C; 80 °C → rt
- 1.2 **Reagents:** Hydrochloric acid
Solvents: Ethyl acetate, Water; 0 °C

Scheme 33 (1 Reaction)

Steps: 1 Yield: 92%



Suppliers (26)

31-614-CAS-41533737

Steps: 1 Yield: 92%

Metal-Free Visible-Light-Induced Hydrodifluoroalkylation of Electron-Deficient Alkenes Using Hantzsch Ester

By: Wang, Juan; et al

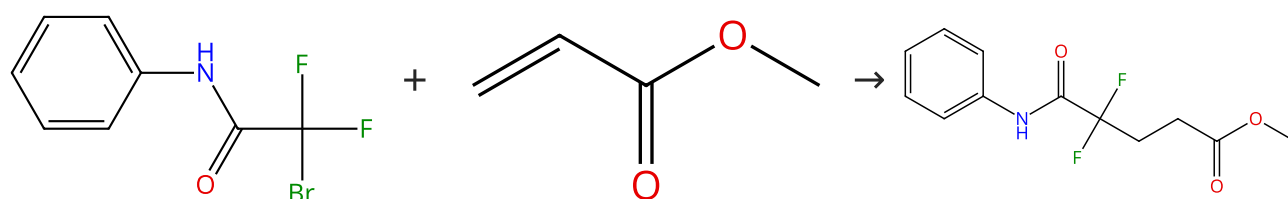
European Journal of Organic Chemistry (2024), 27(38), e202400591.

- 1.1 **Reagents:** Triethylamine, Hantzsch ester
Solvents: 1,4-Dioxane; 12 h, rt

Experimental Protocols

Scheme 34 (1 Reaction)

Steps: 1 Yield: 91%



Suppliers (26)

Suppliers (60)

31-614-CAS-41533724

Steps: 1 Yield: 91%

Metal-Free Visible-Light-Induced Hydrodifluoroalkylation of Electron-Deficient Alkenes Using Hantzsch Ester

By: Wang, Juan; et al

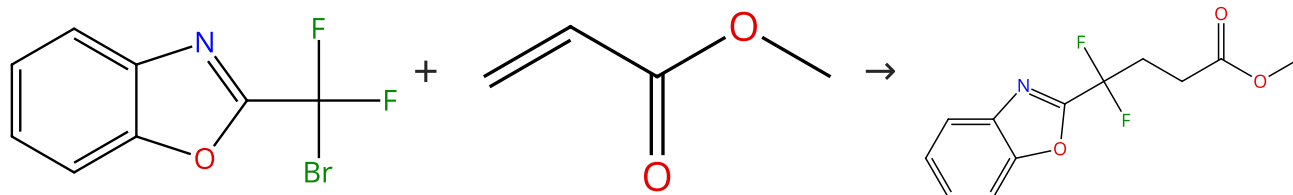
European Journal of Organic Chemistry (2024), 27(38), e202400591.

- 1.1 **Reagents:** Triethylamine, Hantzsch ester
Solvents: 1,4-Dioxane; 12 h, rt

Experimental Protocols

Scheme 35 (1 Reaction)

Steps: 1 Yield: 91%



Suppliers (41)

Suppliers (60)

31-614-CAS-41533744

Steps: 1 Yield: 91%

Metal-Free Visible-Light-Induced Hydrodifluoroalkylation of Electron-Deficient Alkenes Using Hantzsch Ester

By: Wang, Juan; et al

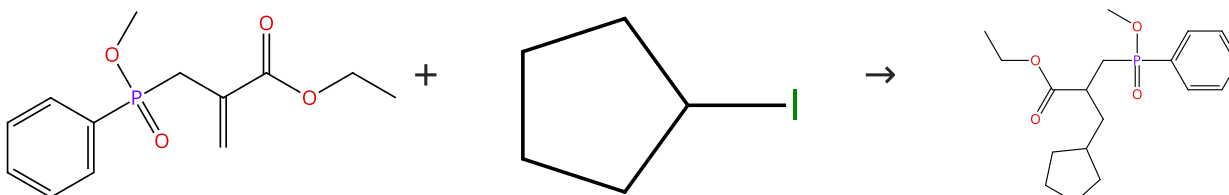
European Journal of Organic Chemistry (2024), 27(38), e202400591.

1.1 **Reagents:** Triethylamine, Hantzsch ester
Solvents: 1,4-Dioxane; 12 h, rt

Experimental Protocols

Scheme 36 (1 Reaction)

Steps: 1 Yield: 91%



Suppliers (61)

31-255-CAS-20187141

Steps: 1 Yield: 91%

Late-stage diversification of phosphinic dehydroalanine pseudopeptides based on a Giese-type radical C-alkylation strategy

By: Voreakos, Kostas; et al

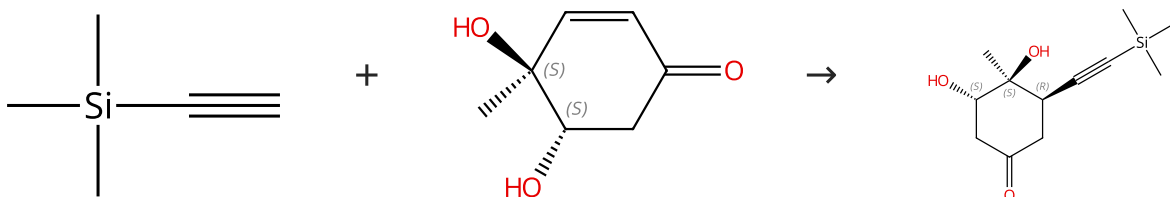
Organic Letters (2019), 21(12), 4397-4401.

1.1 **Reagents:** Sodium cyanoborohydride
Catalysts: Azobisisobutyronitrile
Solvents: Methanol; 0 °C; 1 h, 80 °C; 80 °C → rt

1.2 **Reagents:** Hydrochloric acid
Solvents: Ethyl acetate, Water; rt

Scheme 37 (2 Reactions)

Steps: 1 Yield: 91%



Suppliers (98)

Supplier (1)

31-614-CAS-34511401

Steps: 1 Yield: 91%

Convergent total synthesis of (+)-calcipotriol: a scalable, modular approach to vitamin D analogs

By: Gu, Jieyu; et al

Proceedings of the National Academy of Sciences of the United States of America (2022), 119(18), e2200814119.

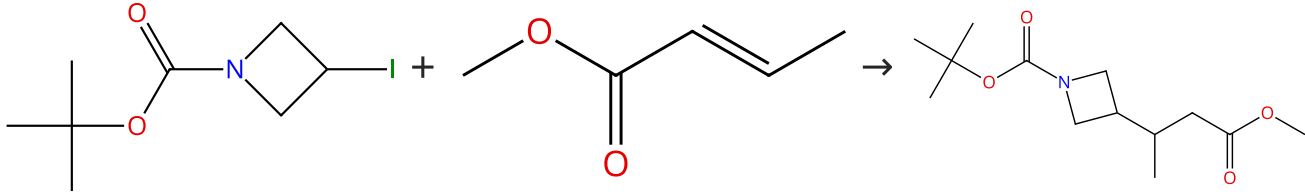
1.1 **Reagents:** Ethylmagnesium bromide
Solvents: Tetrahydrofuran; 0 °C; 10 min, 0 °C → rt; 5 min, rt; 1 h, rt → 50 °C; 50 °C → rt; 5 min, rt; rt → 0 °C

1.2 **Solvents:** Tetrahydrofuran; 15 min, 0 °C; 0 °C → rt; 1 h, rt

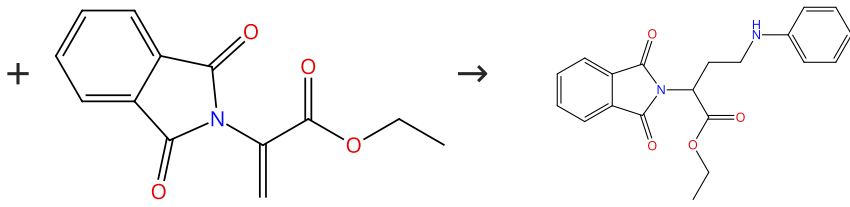
1.3 **Reagents:** Hydrochloric acid
Solvents: Water; rt

Experimental Protocols

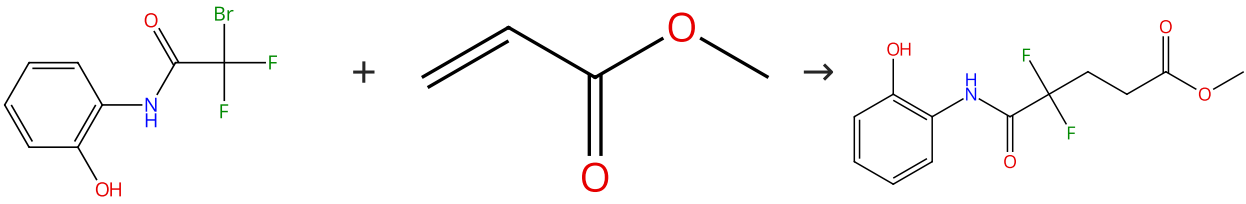
31-614-CAS-31938572 Steps: 1 Yield: 91% 1.1 Reagents: Ethylmagnesium bromide Solvents: Tetrahydrofuran; rt → 0 °C; 10 min, 0 °C; 5 min, rt; 1 h, rt → 50 °C; 50 °C → rt; 5 min, rt; rt → 0 °C 1.2 Solvents: Tetrahydrofuran; 15 min, 0 °C; rt; 1 h, rt 1.3 Reagents: Hydrochloric acid Solvents: Water; rt Experimental Protocols	Convergent total synthesis of (+)-calcipotriol: a scalable, modular approach to vitamin D analogs By: Gu, Jieyu; et al ChemRxiv (2021), 1-17.
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Scheme 38 (1 Reaction) Steps: 1 Yield: 91%   Suppliers (93)  Suppliers (10)  Suppliers (2)	
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31-255-CAS-22353281 Steps: 1 Yield: 91% 1.1 Reagents: Triethylamine Catalysts: 2,4,5,6-Tetra(9<i>H</i>-carbazol-9-yl)isophthalonitrile Solvents: Acetonitrile, Water; 16 h, rt Experimental Protocols	Aminoalkyl radicals as halogen-atom transfer agents for activation of alkyl and aryl halides By: Constantin, Timothee; et al Science (Washington, DC, United States) (2020), 367(6481), 1021-1026.
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Scheme 39 (1 Reaction) Steps: 1 Yield: 90% Multi-component structure image available in CAS SciFinder   Suppliers (5)	
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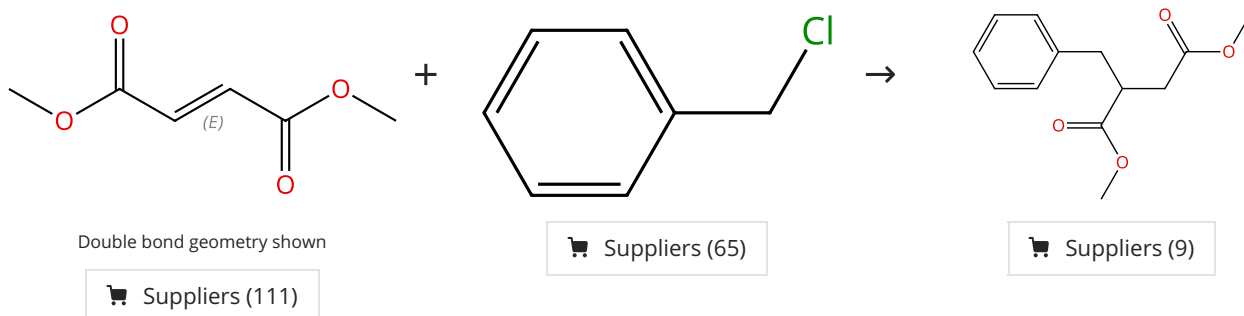
31-614-CAS-32730586 Steps: 1 Yield: 90% 1.1 Catalysts: Iridium(1+), [4,4'-bis(1,1-dimethylethyl)-2,2'-bipyridine-κN¹,κN¹']bis[3,5-difluoro-2-[5-(trifluoromethyl)-2-pyridinyl-κM]phenyl-κC]-, (OC-6-33)-, hexafluorophosphate(1-)(1:1) Solvents: Dimethyl sulfoxide; 24 h, rt Experimental Protocols	Expedient access to N-alkylphthalimides via redox-neutral photocatalysed Giese-type reactions By: Jin, Xiaoping; et al Organic & Biomolecular Chemistry (2022), 20(27), 5377-5382.
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Scheme 40 (1 Reaction) Steps: 1 Yield: 90%   Suppliers (38)  Suppliers (60)	
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31-614-CAS-41533743	Steps: 1 Yield: 90%	Metal-Free Visible-Light-Induced Hydrodifluoroalkylation of Electron-Deficient Alkenes Using Hantzsch Ester
1.1 Reagents: Triethylamine, Hantzsch ester Solvents: 1,4-Dioxane; 12 h, rt		By: Wang, Juan; et al
Experimental Protocols		European Journal of Organic Chemistry (2024), 27(38), e202400591.

Scheme 41 (1 Reaction)

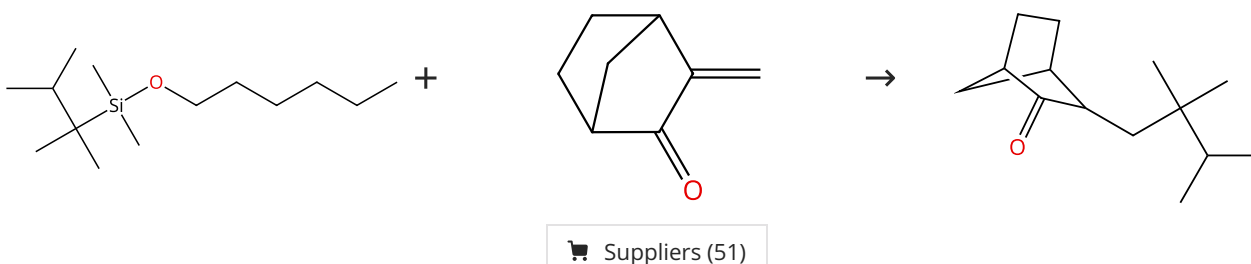
Steps: 1 Yield: 90%



31-255-CAS-20099038	Steps: 1 Yield: 90%	Photochemical generation of radicals from alkyl electrophiles using a nucleophilic organic catalyst
1.1 Reagents: γ -Terpinene, 2,6-Lutidine Catalysts: 1 <i>H</i> -Indole-1-carbodithioic acid, 5-bromo-, potassium salt (1:1) Solvents: Acetonitrile; 24 h, 60 °C		By: Schweitzer-Chaput, Bertrand; et al
		Nature Chemistry (2019), 11(2), 129-135.

Scheme 42 (1 Reaction)

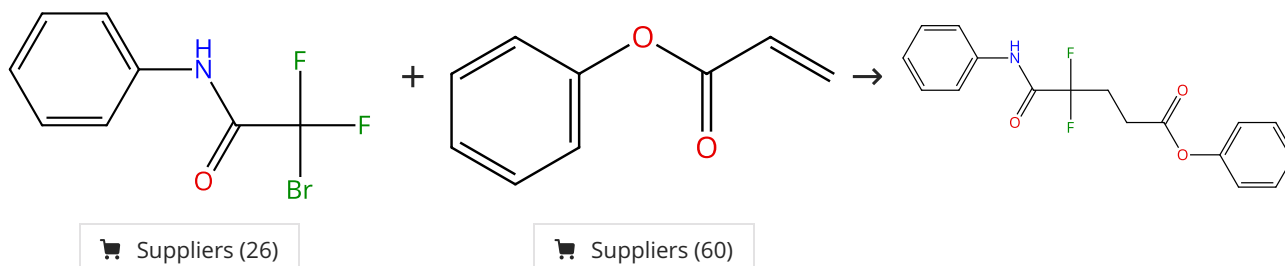
Steps: 1 Yield: 90%



31-614-CAS-42383411	Steps: 1 Yield: 90%	Organophotocatalyzed C-Si Bond Fragmentation Using Silyl Ethers as Radical Precursors
1.1 Catalysts: Biphenyl, Lithium perchlorate, Acridinium, 10-methyl-9-(2,4,6-trimethylphenyl)-, tetrafluoroborate(1-) (1:1) Solvents: Acetonitrile; 24 h, rt		By: Luguera Ruiz, Adrian; et al
Experimental Protocols		Advanced Synthesis & Catalysis (2024), 366(19), 4132-4138.

Scheme 43 (1 Reaction)

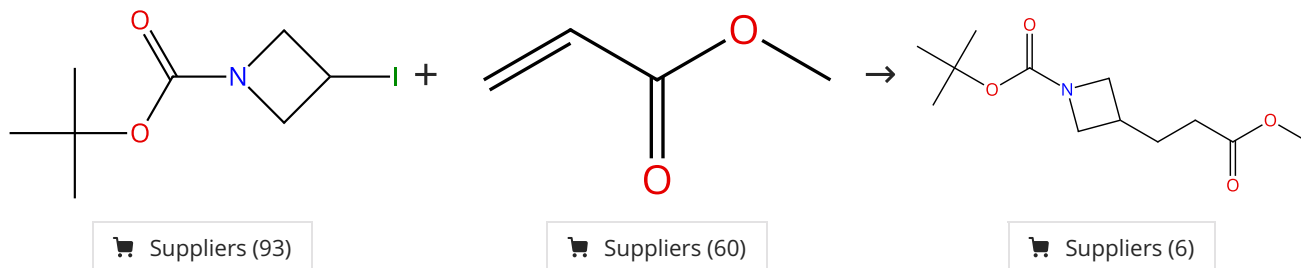
Steps: 1 Yield: 90%



31-614-CAS-41533725	Steps: 1 Yield: 90%	Metal-Free Visible-Light-Induced Hydrodifluoroalkylation of Electron-Deficient Alkenes Using Hantzsch Ester
1.1 Reagents: Triethylamine, Hantzsch ester Solvents: 1,4-Dioxane; 12 h, rt		By: Wang, Juan; et al
Experimental Protocols		European Journal of Organic Chemistry (2024), 27(38), e202400591.

Scheme 44 (1 Reaction)

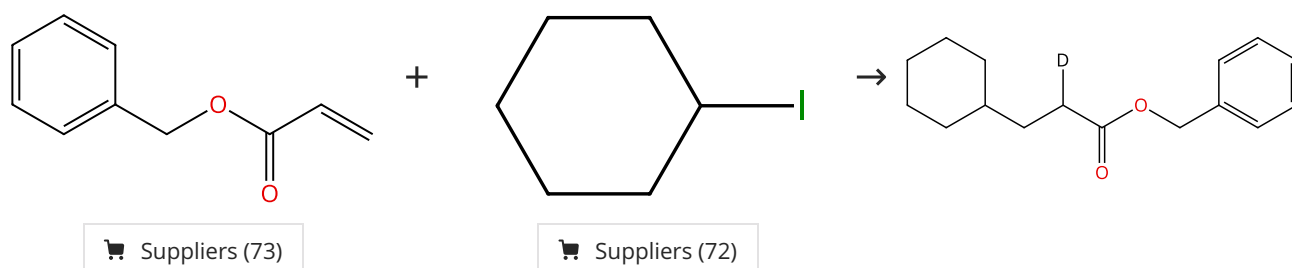
Steps: 1 Yield: 90%



31-255-CAS-22353272	Steps: 1 Yield: 90%	Aminoalkyl radicals as halogen-atom transfer agents for activation of alkyl and aryl halides
1.1 Reagents: Triethylamine Catalysts: 2,4,5,6-Tetra(9 <i>H</i> -carbazol-9-yl)isophthalonitrile Solvents: Acetonitrile, Water; 16 h, rt		By: Constantin, Timothee; et al
Experimental Protocols		Science (Washington, DC, United States) (2020), 367(6481), 1021-1026.

Scheme 45 (1 Reaction)

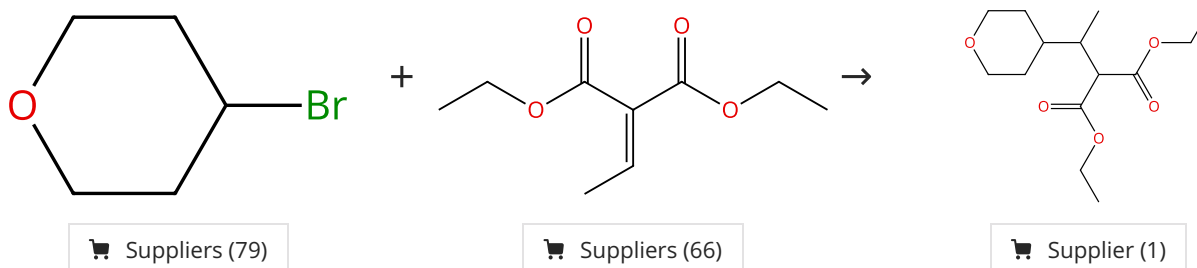
Steps: 1 Yield: 90%



31-116-CAS-20578180	Steps: 1 Yield: 90%	Visible-light-initiated manganese-catalyzed Giese addition of unactivated alkyl iodides to electron-poor olefins
1.1 Reagents: 3,5-Diethyl 1,4-dihydro-4- <i>d</i> -2,6-dimethyl-3,5-pyridine-4- <i>d</i> -dicarboxylate Catalysts: Dimanganese decacarbonyl Solvents: Dimethyl sulfoxide; 24 h, rt		By: Dong, Jianyang; et al
		Chemical Communications (Cambridge, United Kingdom) (2019), 55(78), 11707-11710.

Scheme 46 (1 Reaction)

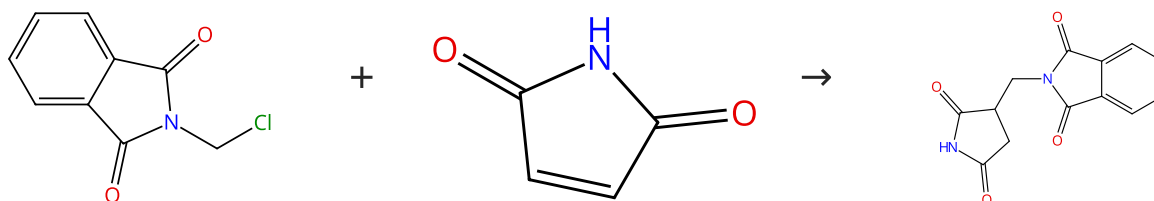
Steps: 1 Yield: 89%



31-255-CAS-19797522	Steps: 1 Yield: 89%	Silyl-mediated photoredox-catalyzed Giese reaction: addition of non-activated alkyl bromides
1.1 Reagents: Sodium carbonate, Tris(trimethylsilyl)silane Catalysts: Iridium(1+), [4,4'-bis(1,1-dimethylethyl)-2,2'-bipyridine- $\kappa N^1, \kappa N^1$]bis[3,5-difluoro-2-[5-(trifluoromethyl)-2-pyridinyl- κN]phenyl- κC]-, (OC-6-33)-, hexafluorophosphate(1-)(1:1) Solvents: Methanol; 1 h, 25 °C		By: ElMarrouni, Abdellatif; et al Chemical Science (2018), 9(32), 6639-6646.
Experimental Protocols		

Scheme 47 (1 Reaction)

Steps: 1 Yield: 89%



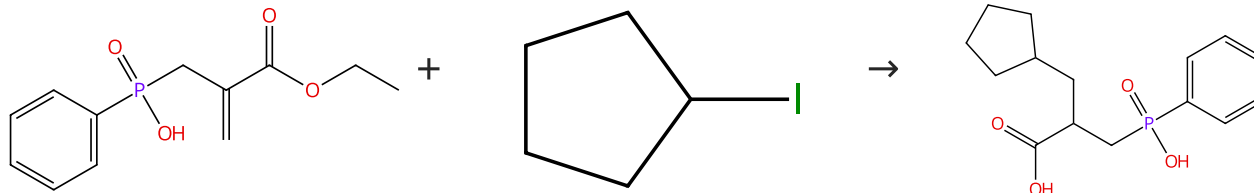
Suppliers (67)

Suppliers (92)

31-255-CAS-20099056	Steps: 1 Yield: 89%	Photochemical generation of radicals from alkyl electrophiles using a nucleophilic organic catalyst
1.1 Reagents: γ -Terpinene, 2,6-Lutidine Catalysts: 1 <i>H</i> -Indole-1-carbodithioic acid, 5-bromo-, potassium salt (1:1) Solvents: Acetonitrile; 24 h, 60 °C		By: Schweitzer-Chaput, Bertrand; et al Nature Chemistry (2019), 11(2), 129-135.

Scheme 48 (1 Reaction)

Steps: 1 Yield: 89%

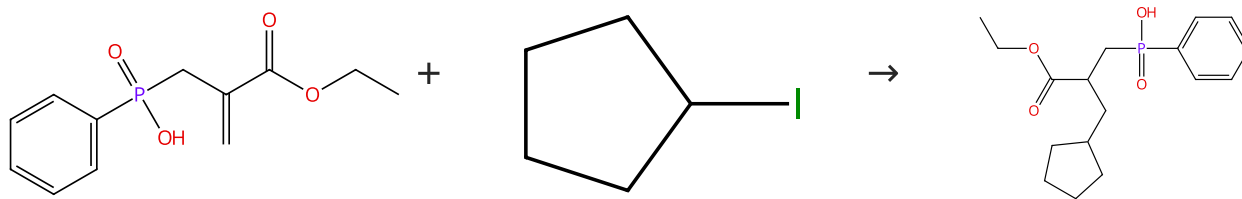


Suppliers (61)

31-255-CAS-20187109	Steps: 1 Yield: 89%	Late-stage diversification of phosphinic dehydroalanine pseudopeptides based on a Giese-type radical C-alkylation strategy
1.1 Reagents: Tributylstannane Catalysts: Triethylborane Solvents: Dichloromethane, Hexane; 5 min, 0 °C; 30 min, 0 °C; 30 min, 0 °C		By: Voreakos, Kostas; et al Organic Letters (2019), 21(12), 4397-4401.
1.2 Reagents: Hydrochloric acid Solvents: Water; 10 min, 0 °C		
1.3 Reagents: Sodium hydroxide Solvents: Ethanol, Water; 24 h, rt		
1.4 Reagents: Hydrochloric acid Solvents: Water; pH 1, rt		

Scheme 49 (1 Reaction)

Steps: 1 Yield: 89%



Suppliers (61)

31-255-CAS-20187149

Steps: 1 Yield: 89%

Late-stage diversification of phosphinic dehydroalanine pseudopeptides based on a Giese-type radical C-alkylation strategy

By: Voreakos, Kostas; et al

Organic Letters (2019), 21(12), 4397-4401.

- 1.1 **Reagents:** Tributylstannane
Catalysts: Triethylborane
Solvents: Dichloromethane, Hexane; 5 min, 0 °C; 30 min, 0 °C; 30 min, 0 °C
- 1.2 **Reagents:** Hydrochloric acid
Solvents: Water; 10 min, 0 °C
- 1.3 **Reagents:** Hydrochloric acid
Solvents: Ethyl acetate, Water; 0 °C